Exam #4:

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Problem 1:

$A \rightarrow B$ (zeroth order)

Assume: one dimensional, isothermal

a) Mass balance:

$$\frac{d^2C_A}{dr^2} + \frac{2}{r}\left(\frac{dC_A}{dr}\right) - \frac{k_0}{D_e}C_A^n = 0$$

$\frac{10}{20}$

Also can be written as:

$$\left(\frac{2D_e}{r}\right)\frac{dC_A}{dr} = k_0 \quad \text{(assuming linear kinetics)}$$

??

Boundary Conditions:

B.C. 1: $C_A(r=R) = C_{AS}$

Assume: B.C. 2: $C_A(r=0) =$ finite ; $\left.\frac{dC_A}{dr}\right|_{r=0} = 0$

$$C_A(r) = \underbrace{A\sinh\left(\sqrt{\tfrac{k_0}{D}}r\right) + B\cosh\left(\sqrt{\tfrac{k_0}{D}}r\right)}_{\text{2 integration constants.}}$$

From MB & B.C. 2:

$$: \left[A\sqrt{\frac{k_0}{D}}\cosh\left(\sqrt{\frac{k_0}{D}}r\right) + B\sqrt{\frac{k_0}{D}}\left(\sinh\sqrt{\frac{k_0}{D}}x\right)\right]_{x=0} = 0$$

$$\Rightarrow A\sqrt{\frac{k_0}{D}}\cos(0) + 0 = 0$$

$$\Rightarrow \boxed{A = 0}$$

$$\text{B.C. 1} = B\cosh\left(\sqrt{\tfrac{k_1}{D}}R\right) = C_{AS}$$

$$B = \frac{C_{AS}}{\cosh\left(\sqrt{\frac{k_0}{D}}R\right)}$$

$$C_A = C_{AS} + \frac{k_0}{6D}(r^2 - R^2)$$

$$\boxed{C_A(r) = \frac{C_{AS}}{\cosh\left(\frac{k_0}{D}R\right)} \cdot \cosh\left(\sqrt{\frac{k_0}{D}}r\right)}$$

B) $\eta = \frac{-r_A \; m_p}{-r_{AS} \; m_{AP}}$ $\quad \phi = \frac{k_0 R^2}{D_e}$

$\eta = 1$

$\frac{4}{20}$

$$\eta = \frac{4\pi R D_e C_{AS}}{k_0 \alpha C_{AS} \frac{4}{3}\pi R^3}(\phi\coth\phi - 1) = \frac{3}{\phi^2}(\phi\coth\phi - 1)$$

$$= \frac{3D_e}{k_0 R^2}\left(\frac{k_0 R^2}{D_e}\coth\left(\frac{k_0 R^2}{D_e}\right) - 1\right)$$

c) $\frac{2D^2C_A}{r\,dr} = r_0$ $\sqrt{\frac{k_0}{DC_{As}}}R \ll 1$

(2/10)

$\frac{dC_A}{dr}\Big|_{r=0} = 0$ ×

d) $r_A = \frac{D \cdot A \cdot (C_R - C_r)}{h}$ ×

(1/10)

problem 2:

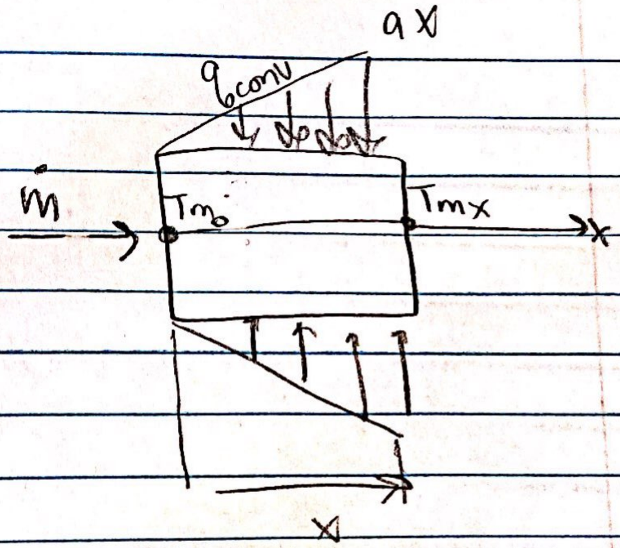
given: $T_o = 27^\circ C$

$\dot{m} = 450 \, kg/h$

$q'(W/m) = \alpha \cdot x$

$\alpha = 20 \, W/m^2$

$C_p = 4.18 \, kJ/kg \cdot K$



* we have a control volume

Do an energy balance: $dq_{conv} = \dot{m} c_p dT$

$$\int_0^x q'_{conv} dx = \dot{m} C_p \int_{T_{m_o}}^{T_{mx+dx}} dT$$

$$\Rightarrow \int_0^x (\alpha x) dx = \dot{m} C_p \int_{T_{m_o}}^{T_{mx}} dT$$

$$20 \left[\frac{x^2}{2} \right]_0^x = \dot{m} C_p (T_{mx} - T_{m_o})$$

$$\Rightarrow \boxed{T_{mx} = T_{m_o} + \frac{10x^2}{\dot{m} C_p}}$$

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Outlet temp for 30m long section:

$$T_{mx} = 27 + \frac{10(30)^2}{\left(\frac{450}{(60)(60)}\right)\left(4180 \frac{J}{kg \cdot K}\right)} = \boxed{44.22^\circ C}$$

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